

1.

$$x_0 = 1; (w_0, w_1, w_2) = (0, 1, -0, 3, 0, 4)$$

$$(x_1, x_2) = (1, -1)$$

$$\text{Target } t = -1$$

$$\text{LR } \eta = 0,2$$

$$\text{Net input } z = w_0 x_0 + w_1 x_1 + w_2 x_2$$

$$z = (0, 1) \cdot 1 + (0, 3) \cdot 1 + (0, 4) \cdot (-1)$$

$$= -0,6$$

$$\text{Output } y = \begin{cases} +1 & z \geq 0 \\ -1 & z < 0 \end{cases} \Rightarrow y = -1$$

$\Rightarrow$ Perception correct $\rightarrow$ No weight update needed

2. $z \in \{-3, -0,5, 0, 2\}$

$$\text{ReLU}(z) = \max(0, z)$$

$$\text{Sigmoid } \sigma(z) = \frac{1}{1 + e^{-z}}$$

z	ReLU
-3	0
-0,5	0
0	0
2	2

z	Sigmoid
-3	0,047
-0,5	0,378
0	0,5
2	0,881

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Saturation:

Sig $\uparrow$ when

Sigmoid saturate when z large. $z \gg 0 \rightarrow$ output approach 1.

$z \ll 0 \rightarrow$ output approaches 0.

$\sigma'(z) = \sigma(z)(1 - \sigma(z))$ approaches 0 $\rightarrow$ gradient small.

$\rightarrow$ Vanishing gradient

ReLU doesn't saturate at positive.

$$\text{ReLU}'(z) = 1, \quad z > 0$$

$\rightarrow$ gradient pass better when activation positive.

Zero gradient:

$$\text{ReLU}'(z) = 0, \quad z \leq 0$$

If neuron continuously receives 0 $\rightarrow$ stop learning as gradient = 0

$\rightarrow$ dying ReLU

Sigmoid has gradient $\neq 0$ at infinite z , but small when z saturates.

Output range:

$$\text{ReLU} \in [0, \infty)$$

$$\text{Sigm} \in (0, 1)$$

tanh better when hidden representations are zero centered.

* $\tanh \in (-1, 1)$ when sigmoid always positive
 $\rightarrow$ when features need to represent both positive and negative values $\rightarrow$ $\tanh$ better

3. $x = 1$, $w = 0,6$, $v = -0,7$

Bias = 0, target $t = 0$, loss $L = \frac{1}{2} (1 - 0)^2$

1. Forward thru hidden unit

$$z_h = wx = 0,6 \cdot 1 = 0,6$$

$$h \approx \sigma(0,6) = 0,646$$

2. Forward thru output

$$z_o = vh = (-0,7)(0,646) = -0,452$$

$$o = \sigma(-0,452)$$

$$= 0,389$$

3. Error signal:

$$\frac{\partial L}{\partial o} = o - t$$

$$\delta_o = (o - t)o(1 - o)$$

$$\rightarrow = (0,389 - 0)(0,389)(1 - 0,389)$$

$$\Rightarrow \frac{\partial o}{\partial z_o} = o(1 - o)$$

$$= \delta_o \approx + 0,092$$

$\rightarrow$ positive

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4. Gradient on v ✓

$$\frac{\partial L}{\partial v} = \delta_h \approx 0,092 \cdot (0,646) = +0,060$$

$$v_{\text{new}} = v - \eta \frac{\partial L}{\partial v} \rightarrow v \text{ decrease because } \frac{\partial L}{\partial v} \text{ positive}$$

5. Gradient on w .

$$\left. \begin{array}{l} \delta_h = \delta_o v h (1-h) \\ \delta_o > 0 \\ v < 0 \\ h(1-h) > 0 \end{array} \right\} \rightarrow \delta_h < 0$$

$$\left. \begin{array}{l} \frac{\partial L}{\partial w} = \delta_h x \\ x = 1 > 0 \end{array} \right\} \rightarrow \frac{\partial L}{\partial w} < 0$$

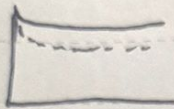
$$w_{\text{new}} = w - \eta \frac{\partial L}{\partial w}$$

$\rightarrow w$ increases.

$$\frac{\partial L}{\partial w} \approx -0,0148$$

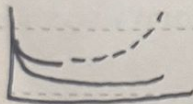
$\rightarrow v \downarrow, w \uparrow$

4. underfit - train -- validation



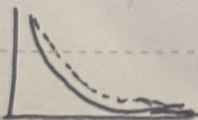
→ need increase complexity.

overfit



→ early stop, reduce complexity

balance



→ no need intervention

5.

I will compare two MLPs with hidden widths of 16 and 64 units, while keeping all other settings fixed. I expect the 64 unit model to achieve higher training score because of greater capacity. Its validation score ~~should~~ ^{may} also improve if smaller model underfits, though the improvement may be limited or reversed if larger model overfits. I also expect the 64 to take longer to train because it has more parameters.

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	16-bit	64-bit
train macro-f1	0,9956	1,0000
val macro-f1	0,9700	0,9738
runtime	6,66s	12,45s
converges	252	226

→ Correct prediction, bigger model provide small improvement, but more computation clearly